

# Chapter 1 – Overview

DATA 335 – Statistical Modeling – Winter 2026

# Statistical inference

This is a course about *statistical inference*.

## Statistical inference

---

Article [Talk](#)

---

From Wikipedia, the free encyclopedia

*Not to be confused with [Statistical interference](#).*

**Statistical inference** is the process of using [data analysis](#) to infer properties of an underlying [probability distribution](#).<sup>[1]</sup> **Inferential statistical analysis** infers properties of a [population](#), for example by **testing hypotheses** and deriving estimates. It is assumed that the observed data set is [sampled](#) from a larger population.

[1] Upton, G., Cook, I. (2008) Oxford Dictionary of Statistics, OUP

# Do smoking moms have smaller babies?

16 babies of nonsmoking moms, 16 babies of smoking moms

